

MEME16203 Linear Models**Assignment 3****UNIVERSITI TUNKU ABDUL RAHMAN**

Faculty:	FES	Unit Code:	MEME16203
Course:	MAC	Unit Title:	Linear Models
Year:	1,2	Lecturer:	Dr Yong Chin Khian
Session:	202606		
Due by:			

Q1. Suppose $\mathbf{M}$ is an $m \times n$ matrix of rank r . Suppose $\mathbf{z}$ is an m -dimensional standard multivariate normal random vector.

- (a) Provide an expression involving $\mathbf{M}$ and $\mathbf{z}$ that has a central chi-squared distribution with r degrees of freedom.
- (b) Show that the expression involving $\mathbf{M}$ and $\mathbf{z}$ that you provided in part (a) has a central chi-squared distribution with r degrees of freedom.
- (c) Let A be the random variable involving $\mathbf{M}$ and $\mathbf{z}$ that you defined in part (a). Let $B = \mathbf{z}^T \mathbf{z}$. Provide an expression involving a and b that has a central F distribution with r numerator degrees of freedom and $m - r$ denominator degrees of freedom.
- (d) Show that the expression you provided in part (c) has a central F distribution with r numerator degrees of freedom and $m - r$ denominator degrees of freedom.

[Total:20 marks]

Q2. Consider the model $Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$, where $i = 1, 2, 3$, $j = 1, 2, 3$, and μ , α_1 , α_2 , α_3 , are unknown parameters. Let $\epsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$, where σ^2 is unknown.

- (a) Verify that $\tau = 4\mu + 3\alpha_1 + 7\alpha_2 + 4\alpha_3$ is an estimable function.
- (b) Write down a formula for $\hat{\tau}$, the BLUE for τ .
- (c) Determine the distribution of $\frac{\hat{\tau}^2}{25\sigma^2}$.
- (d) Determine the distribution of $S^2 = \sum_{i=1}^3 \sum_{j=1}^3 (Y_{ij} - \bar{Y}_i)^2$.
- (e) Show that $F = \frac{c\hat{\tau}^2}{S^2}$, where c is a constant, has central F -distribution when $\tau = 0$. Report c .

[Total:20 marks]

Q3. Consider the model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i(1)$$

where $\epsilon \sim NID(0, \sigma^2)$. This notation means that the random errors (and the observations) have normal distributions and satisfy the Gauss-Markov property.

MEME16203 Linear Models

Define

$$\mathbf{X} = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix} \text{ and } \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \text{ and } \mathbf{P}_X = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \text{ and } \mathbf{P}_1 = \mathbf{1}(\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T$$

- (a) What are the distribution of the quadratic forms $\frac{1}{\sigma^2} \mathbf{Y}^T (\mathbf{P}_X - \mathbf{P}_1) \mathbf{Y}$ and $\frac{1}{\sigma^2} \mathbf{Y}^T (\mathbf{I} - \mathbf{P}_X) \mathbf{Y}$.
- (b) Derive the distribution of $F = \frac{(n-2) \mathbf{Y}^T (\mathbf{P}_X - \mathbf{P}_1) \mathbf{Y}}{\mathbf{Y}^T (\mathbf{I} - \mathbf{P}_X) \mathbf{Y}}$. Report degrees of freedom and a formula for the noncentrality parameter.
- (c) What is the null hypothesis associated with the F statistic in part (b)? Justify your answer by showing that the noncentrality parameter in part (b) is zero if and only if the null hypothesis is true.

[Total:20 marks]

Q4. Consider the model

$$y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk}$$

$$i = 1, 2; j = 1, 2; k = 1, 2$$

where $\epsilon_{ijk} \sim NID(0, \sigma^2)$, α_i is associated with the i -th level of factor A , β_j is associated with the j -th level of factor B , and γ_{ij} is an interaction parameter.

- (a) Define $SSE = \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^2 (y_{ijk} - \bar{y}_{ij\bullet})^2$, where $\bar{y}_{ij\bullet} = \frac{1}{2}(y_{ij1} + y_{ij2})$. Show that $\frac{SSE}{\sigma^2}$ has a chi-squares distribution. States the degrees of freedom.

- (b) Consider the estimator

$$\hat{C} = \bar{y}_{2\bullet\bullet} - \bar{y}_{1\bullet\bullet},$$

where

$$\bar{y}_{i\bullet\bullet} = \frac{1}{4} \sum_{j=1}^2 \sum_{k=1}^2 y_{ijk}.$$

Show that

$$F = \frac{k(\hat{C})^2}{SSE}$$

has an F-distribution for some constant k . Report the value of k and the degrees of freedom for the F-distribution.

[Total:20 marks]

- Q5. A study of the effect of temperature $\mathbf{x}_1$ and type of fabrics $\mathbf{x}_2$ on percent shrinkage in dyeing fabrics was made in a completely randomized design. The data $\mathbf{y}$ are the percent shrinkage of two replicate fabrics pieces dried at each of the three temperatures. The data are shown below. In this table, $\mathbf{x}_2 = 1$ if Fabric 1 was used and $\mathbf{x}_2 = -1$ if Fabric 2 was used.

MEME16203Linear Models

Percent shrinkage, $\mathbf{y}$	Temperature ($^{\circ}F$), $\mathbf{x}_1$	Fabric, $\mathbf{x}_2$
$y_1, \dots, y_4$	210	1
$y_5, \dots, y_8$	215	1
$y_9, \dots, y_{12}$	225	1
$y_{13}, \dots, y_{16}$	210	-1
$y_{17}, \dots, y_{20}$	215	-1
$y_{21}, \dots, y_{24}$	225	-1

Consider the model

$$y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

where ϵ_i is independently and identically distributed $N(0, \sigma^2)$ for $i = 1, 2, \dots, 24$.

- (a) Let SSE denote the sum of squared residuals for this model. Assume $\mathbf{y} \sim N(\mathbf{x}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, where $\boldsymbol{\beta} = (\beta_1, \beta)^T$. Show that $\frac{1}{\sigma^2}SSE$ has a chi-square distribution.
- (b) Show that $\frac{(\sum_{i=1}^{12} y_i - \sum_{i=13}^{24} y_i)^2}{v\sigma^2}$ has a chi-square distribution. Report v and its degrees of freedom.
- (c) To test for a fabric effect, the researchers propose the following test statistic

$$F = \frac{(\sum_{i=1}^{12} y_i - \sum_{i=13}^{24} y_i)^2}{22SSE}$$

Show that this statistic has an F -distribution. Report degrees of freedom.

- (d) With respect to $\boldsymbol{\beta} = (\beta_1 \ \beta_2)^T$, describe the null hypothesis that can be tested with the F-test in Part (c). What is the alternative hypothesis?

[Total:20 marks]